

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:		Duration:	60 minutes

Code of Conduct

*I **RAGUL SG – 192525007** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the Student

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1. INTRODUCTION AND PROBLEM OVERVIEW

1.1 PROBLEM STATEMENT

The assigned problem statement requires the design and documentation of comprehensive test cases for the FoodieConnect Online Food Ordering System — a software application that allows registered customers to browse restaurant menus, add items to a shopping cart, place orders, make secure online payments, and track the real-time status of their deliveries, while restaurant administrators manage the menu, accept or reject incoming orders, and view sales reports. Handling all of this manually is slow, error-prone, and does not scale as orders grow. The assignment therefore asks for test cases that verify the correctness, functionality, reliability, and usability of the system, covering normal, boundary, invalid, and exceptional conditions, and applying techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition Testing wherever applicable.

1.2 SYSTEM OVERVIEW

The implemented solution is a three-tier web application. The customer-facing front-end (responsive web and mobile interface) communicates with the application server, which interacts with the MySQL database and the Razorpay payment gateway. The system supports three roles: the Customer, who browses, orders, pays, and tracks; the Restaurant Administrator, who controls the menu and order pipeline; and the Delivery Executive, who updates delivery status. The high-level architecture of the system is shown below.

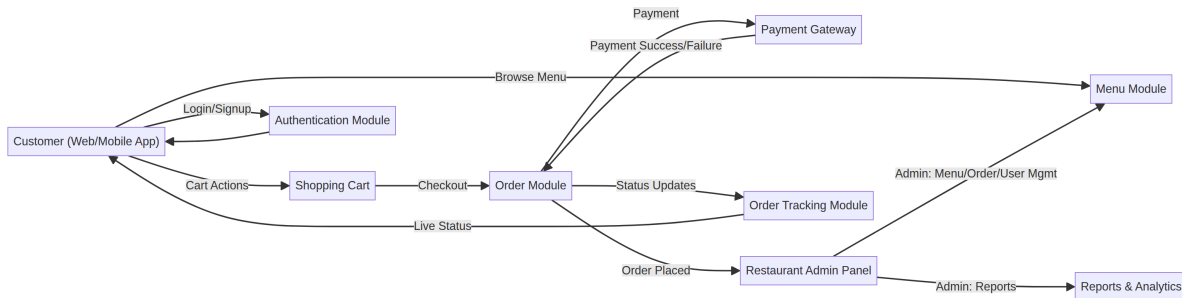


Figure 1: High-level functional architecture of the FoodieConnect system.

1.3 KEY FUNCTIONALITIES

The test suite must verify six key functional areas of the system. The Authentication module handles user registration (name, e-mail, mobile, password) and login, with the account locked after five consecutive failed attempts. The Menu module lists all items together with their availability and supports search by keyword along with filtering by category and price. The Cart module allows items to be added, updated, and removed, and must compute order totals correctly, including tax and delivery charges. The Payment module collects the delivery address, applies the delivery-fee slabs, and processes a secure gateway payment that can be retried on failure. The Order module shows live order status as it advances from Confirmed through Preparing and Out for Delivery to Delivered, and allows cancellation within a two-minute grace period after payment. Finally, the Admin panel lets the restaurant administrator manage menu items, accept or reject orders, and generate date-range sales reports.

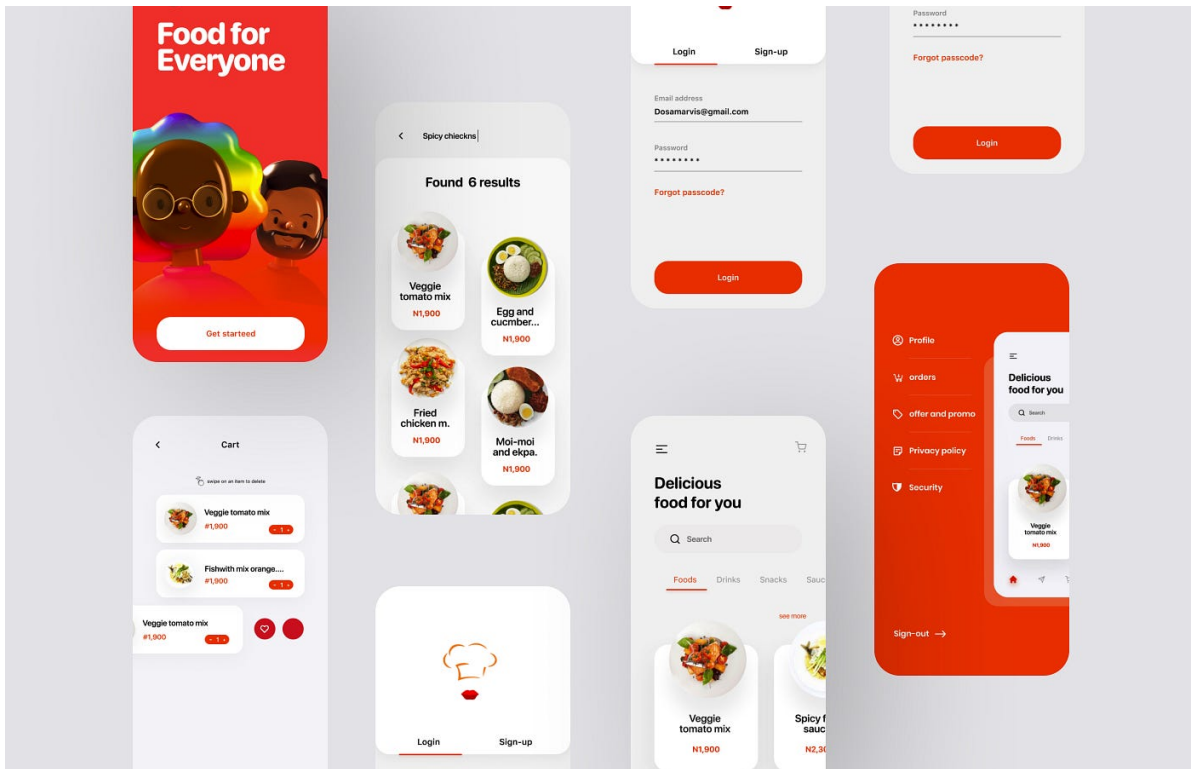


Figure 2: Representative screens of the system under test.

2. REQUIREMENTS TO BE TESTED

Every test case in this report is traceable to a stable requirement identifier. Sixteen functional requirements (FR-01 to FR-16) and eight non-functional requirements (NFR-01 to NFR-08) were identified and are listed below; their coverage is demonstrated in Section 5.

2.1 FUNCTIONAL REQUIREMENTS

In the Authentication module, FR-01 requires that a user can register with name, e-mail, mobile number, and password, and FR-02 requires login with valid credentials together with account lockout after five failed attempts. In the Menu module, FR-03 requires the complete menu to be displayed with each item's name, description, price, and availability, while FR-04 requires search by keyword and filtering by category and price range. In the Cart module, FR-05 covers adding items, updating quantities, and removing items, and FR-06 requires that totals are computed correctly including tax and delivery charges. At checkout, FR-07 requires a delivery address and payment-method selection, and FR-08 requires online payments to be processed with the transaction status recorded. For orders, FR-09 requires order placement with a confirmation showing a unique Order ID, and FR-10 requires live order status through every delivery state. On cancellation, FR-11 allows free cancellation within two minutes of payment and routes later cancellations through the refund workflow, while FR-12 requires the refund to be credited to the original payment source within the SLA. The Admin module requires that administrators add, update, and deactivate menu items (FR-13), accept or reject orders and update their status (FR-14), and generate sales reports for a selected date range (FR-15). Finally, FR-16 requires e-mail/SMS notifications at order, dispatch, and delivery.

2.2 NON-FUNCTIONAL REQUIREMENTS

The eight non-functional requirements span six quality categories. On the performance side, NFR-01 requires the home page to load within three seconds on both Wi-Fi and throttled 4G, and NFR-02 requires search results to return within two

seconds for a 500-item catalog. NFR-03 demands an availability of at least 99.5% per month. Under security, NFR-04 requires passwords to be stored as salted hashes and never written to plain-text logs, while NFR-05 requires payment traffic to travel over TLS 1.2+ with card data remaining exclusively with the gateway. On usability, NFR-06 requires first-time users to complete login, add-to-cart, and checkout unaided; under compatibility, NFR-07 requires correct rendering at 360 px (mobile) and 1920 px (desktop); and on reliability, NFR-08 requires the system to sustain 200 concurrent users with a transaction failure rate under 1%.

3. TEST SCENARIOS AND DESIGN TECHNIQUES

3.1 TEST SCENARIOS

Forty test scenarios (SC-AUTH-01 to SC-NFR-06) were derived module-by-module from the requirements, with each scenario mapped to the requirement or requirements it exercises. For authentication, six scenarios cover valid registration (SC-AUTH-01), rejection of missing or invalid fields (SC-AUTH-02), duplicate e-mail detection (SC-AUTH-03), login with valid credentials (SC-AUTH-04), exact behaviour at the fifth failed login (SC-AUTH-05), and the password-reset flow (SC-AUTH-06).

For the menu, four scenarios verify complete display with availability (SC-MENU-01), matching and non-matching keyword search (SC-MENU-02), category and price filters (SC-MENU-03), and the prevention of adding unavailable items to the cart (SC-MENU-04). Four cart scenarios verify item addition and count updates (SC-CART-01), quantity change recomputation of line totals (SC-CART-02), removal and total updates (SC-CART-03), and cart totals at boundary quantities and the minimum order (SC-CART-04).

The payment module contributes six scenarios covering checkout with a valid address and payment method (SC-PAY-01), successful payment placing the order (SC-PAY-02), graceful failure handling with cart preservation (SC-PAY-03), blocking of orders below the minimum value (SC-PAY-04), delivery-fee changes across the distance slabs (SC-PAY-05), and gateway timeout handling (SC-PAY-06). For orders, five scenarios verify confirmation with a unique Order ID (SC-ORD-01), status transitions from Confirmed to Delivered (SC-ORD-02), rejection of illegal transitions (SC-ORD-03), cancellation inside the two-minute grace period (SC-ORD-04), and the refund workflow after the grace period (SC-ORD-05).

The admin module is covered by five scenarios: adding a valid menu item (SC-ADM-01), blocking invalid item data (SC-ADM-02), order acceptance and rejection outcomes (SC-ADM-03), sales-report correctness with the date-range limit (SC-ADM-04), and denial of admin routes to customers (SC-ADM-05). Finally, six non-functional scenarios verify page-load and search response times (SC-NFR-01), load reliability under 200 concurrent users (SC-NFR-02), password storage and injection resistance (SC-NFR-03), TLS transport and card-data isolation (SC-NFR-04), task completion by first-time users (SC-NFR-05), and rendering across viewports (SC-NFR-06).

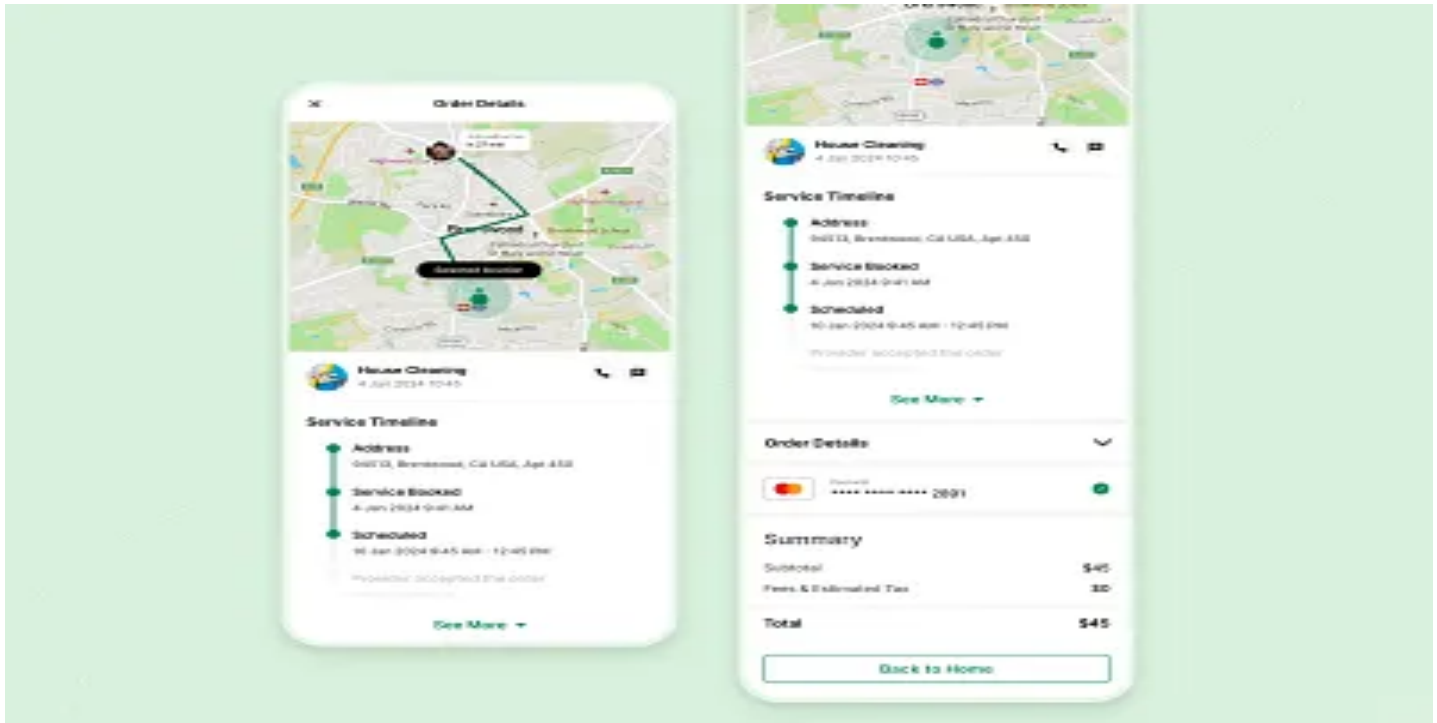


Figure 3: Live order tracking screen exercising SC-ORD scenarios.

3.2 TEST DESIGN TECHNIQUES APPLIED

Four formal test design techniques were applied. Equivalence Partitioning divides each input domain into one valid and several invalid partitions and is used for registration fields, search keywords, and payment methods. Boundary Value Analysis targets the edges of numeric ranges — password length (8–20), quantities (1–99), the Rs.99 minimum order, the 5 km fee slab, and the 120-second cancellation window. Decision Table Testing enumerates the delivery-fee and order-eligibility rules that depend on order value, distance, and payment outcome simultaneously. State Transition Testing verifies every valid transition of the order lifecycle and rejects illegal ones.

As a worked example, the password field of registration (FR-01) yields two equivalence partitions: the valid partition of correctly formatted passwords of 8–20 characters (represented by "Foodie@123", which is accepted) and the invalid partition of passwords that are too short or contain no digit (represented by "Abc1" and "foodiepassword", both rejected with an inline error). Boundary Value Analysis then targets the numeric edges: the minimum length 8 and its predecessor 7 ("Food@123" accepted; "Food@12" rejected) and the maximum length 20 and its successor 21 (a twenty-character valid password accepted; the same password plus one character rejected). The decision table for delivery fees follows this discussion.

Order value $\geq$ Rs.99	Yes	Yes	No	Yes
Distance $\leq$ 5 km	Yes	No	Don't care	Don't care
Payment successful	Yes	Yes	Don't care	No
Allow order	Yes	Yes	No	No
Delivery fee	Rs.30	Rs.50	Not applicable	Not applicable
Block with error	No	No	Yes ("Minimum Rs.99")	No

Payment failure screen	No	No	No	Yes (retry offered)
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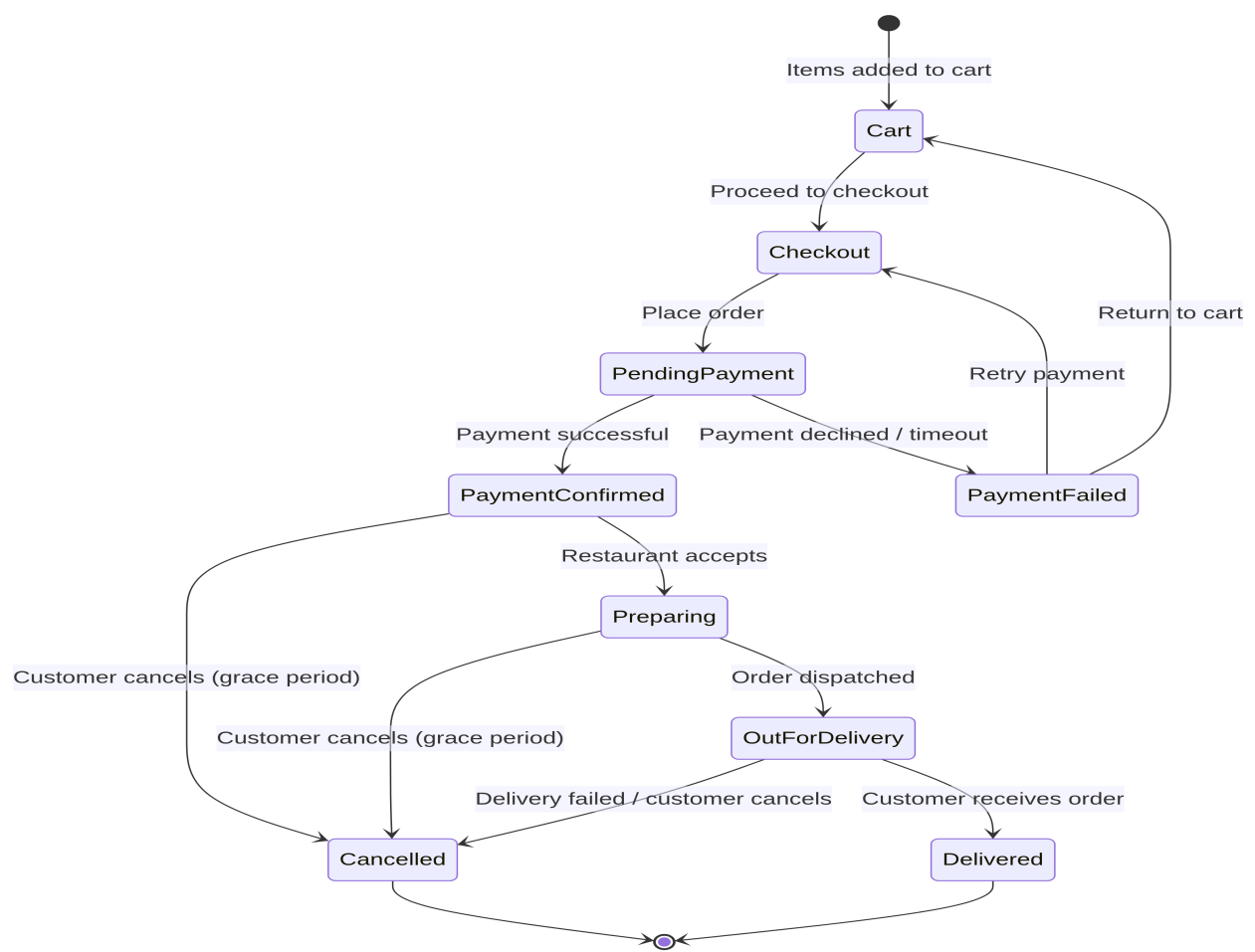


Figure 4: State transition diagram of the order lifecycle.

4. DETAILED TEST CASES

The detailed test cases below cover normal, boundary, invalid, and exceptional conditions. Each test case carries a unique ID, the scenario it belongs to, its type, the test steps, the concrete test data, the expected result, the actual result recorded during execution, and the Pass/Fail status.

4.1 AUTHENTICATION TEST CASES

TC-AUTH-01	SC-AUTH-01	Normal	Open registration; fill fields; click Create Account	Rahul Sharma, rahul.sharma@test.com, +91 98765 43210, Rahul@2026	Account created; confirmation e-mail sent; redirected to login	As expected	Pass
TC-AUTH-02	SC-AUTH-02	Invalid	Leave e-mail blank; short password; 3-digit mobile;	E-mail blank, password Abc1, mobile 123	Inline errors shown; form not submitted	All three errors shown	Pass

			submit				
TC-AUTH-03	SC-AUTH-03	Invalid	Register with an existing e-mail	admin@foodieconnect.test	Rejected with “already registered” message	Rejected as expected	Pass
TC-AUTH-04	SC-AUTH-04	Normal	Log in with valid credentials	rahul.sharma@test.com / Rahul@2026	Login succeeds; session cookie set	Login succeeded	Pass
TC-AUTH-05	SC-AUTH-05	Boundary	Attempt login with wrong password 5 times; try a 6th time	5 wrong passwords, then correct one	Account locked after 5th failure; 6th attempt blocked	Locked at exactly the 5th failure	Pass
TC-AUTH-06	SC-AUTH-06	Exceptional	Request password reset for registered and unregistered e-mails	rahul.sharma@test.com; stranger@nowhere.test	Reset link sent only to the registered e-mail	Registered e-mail received the link	Pass

4.2 MENU, CART AND CHECKOUT TEST CASES

TC-MENU-01	SC-MENU-01	Normal	Open home page; scroll the menu listing	Default URL	All 120 items with name, price, and availability badge	All items displayed	Pass
TC-MENU-02	SC-MENU-02	Normal	Search “pizza”; then search nonsense keyword	pizza; xyzabc	Pizza items returned; zero-match message for nonsense	Both outcomes correct	Pass
TC-MENU-03	SC-MENU-03	Invalid	Filter veg-only and Rs.150–200	veg, Rs.150–200	Only matching veg items shown (Paneer Biryani)	Filtered list correct	Pass

					Rs.199)		
TC-CART-01	SC-CART-01	Normal	Add Margherita Pizza; open cart	Pizza Rs.149	Cart shows 1 item	Cart updated	Pass
TC-CART-02	SC-CART-02	Boundary	Increase quantity to 99; try 100; decrease to 0	Quantity 99 / 100 / 0	99 allowed; 100 rejected; 0 removes item	Boundary enforced	Pass
TC-CART-03	SC-CART-03	Normal	Add two items; remove one; verify totals	Biryani Rs.199, Drink Rs.40	Total reduces by exactly Rs.40	Totals correct	Pass
TC-CART-04	SC-CART-04	Boundary	Build carts of Rs.98 and Rs.99; check out each	Rs.98 vs Rs.99	Rs.98 blocked; Rs.99 proceeds with Rs.30 fee	Minimum enforced at Rs.99	Pass
TC-PAY-01	SC-PAY-01	Normal	Add Biryani; enter MG Road address (3.2 km); select UPI	3.2 km address, UPI	Summary Rs.199 + 5% tax + Rs.30 = Rs.238.95; payment sheet opens	Correct slab and total	Pass
TC-PAY-02	SC-PAY-02	Normal	Pay with the valid test card; confirm	4111 1111 1111, 12/28, CVV 123	Payment succeeds; Order ID generated; status Confirmed	Order ORD-10487 placed	Pass
TC-PAY-03	SC-PAY-03	Invalid	Pay with declining card; retry with valid	4000 0000 0000 0002; retry	Failure screen; cart preserved;	Handled gracefully	Pass

			card	valid	retry succeeds		
TC-PAY-04	SC-PAY-04	Boundary	Order at 5.0 km, then at 5.1 km	5.0 km vs 5.1 km	Fee Rs.30 then Rs.50	Fee changes at boundary	Pass
TC-PAY-05	SC-PAY-05	Exceptional	Inject 45 s gateway delay during payment	Latency 45 s	Loading indicator; no duplicate charge; order settles correctly	Handled without error	Pass

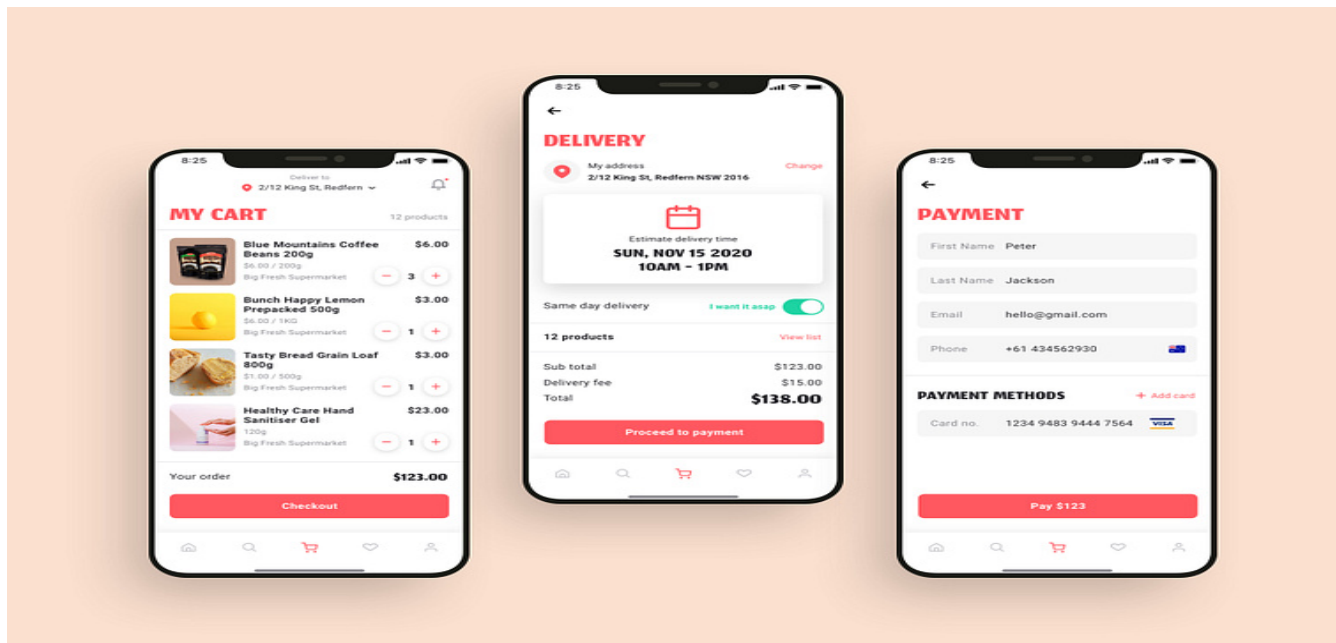


Figure 5: Checkout summary and payment confirmation screens.

4.3 ORDER TRACKING, CANCELLATION AND ADMIN TEST CASES

TC-ORD-01	SC-ORD-01	Normal	Complete payment; view confirmation	Order ORD-10487	Order ID, item list, total, and ETA displayed	Confirmation correct	Pass
TC-ORD-02	SC-ORD-02	Normal	Admin advances status through all four states; customer watches	Confirmed → Preparing → Out for Delivery → Delivered	Customer page updates with timestamps; e-mail/SMS at each step	All transitions visible	Pass

TC-ORD-03	SC-ORD-03	Invalid	Attempt Preparing→Delivered and Delivered→Preparing via API	Illegal transitions	Both rejected with HTTP 409; status unchanged	Blocked as expected	Pass
TC-ORD-04	SC-ORD-04	Boundary	Cancel at 119 s after payment; cancel again at 121 s	t = 119 s; t = 121 s	Full refund at 119 s; refund-request workflow at 121 s	Grace window exact at 120 s	Pass
TC-ORD-05	SC-ORD-05	Normal	Raise refund for delivered order; admin approves; verify credit	Order ORD-10402, Rs.238.95	Refund credited within SLA; history shows Refunded	Refund credited	Pass
TC-ADM-01	SC-ADM-01	Normal	Admin adds valid menu item	Veg Burger, Rs.129, Snacks, veg	Item appears on customer menu immediately	Added and visible	Pass
TC-ADM-02	SC-ADM-02	Invalid	Add item with blank name and price –10	Name blank; price –10	Save blocked with validation errors	Not saved; errors shown	Pass
TC-ADM-03	SC-ADM-03	Normal	Accept ORD-10490; reject ORD-10491	Two queued orders	Accepted moves to Preparing; rejected cancels with refund	Both outcomes verified	Pass
TC-ADM-04	SC-ADM-04	Boundary	Report for 1 day; report for 366 days	Ranges 1 and 366 days	Day report correct; 366-day request rejected (max 365)	Boundary enforced	Pass
TC-ADM-05	SC-ADM-05	Invalid	Customer navigates to /admin	Customer token	HTTP 403; redirect; no admin data exposed	Access denied	Pass

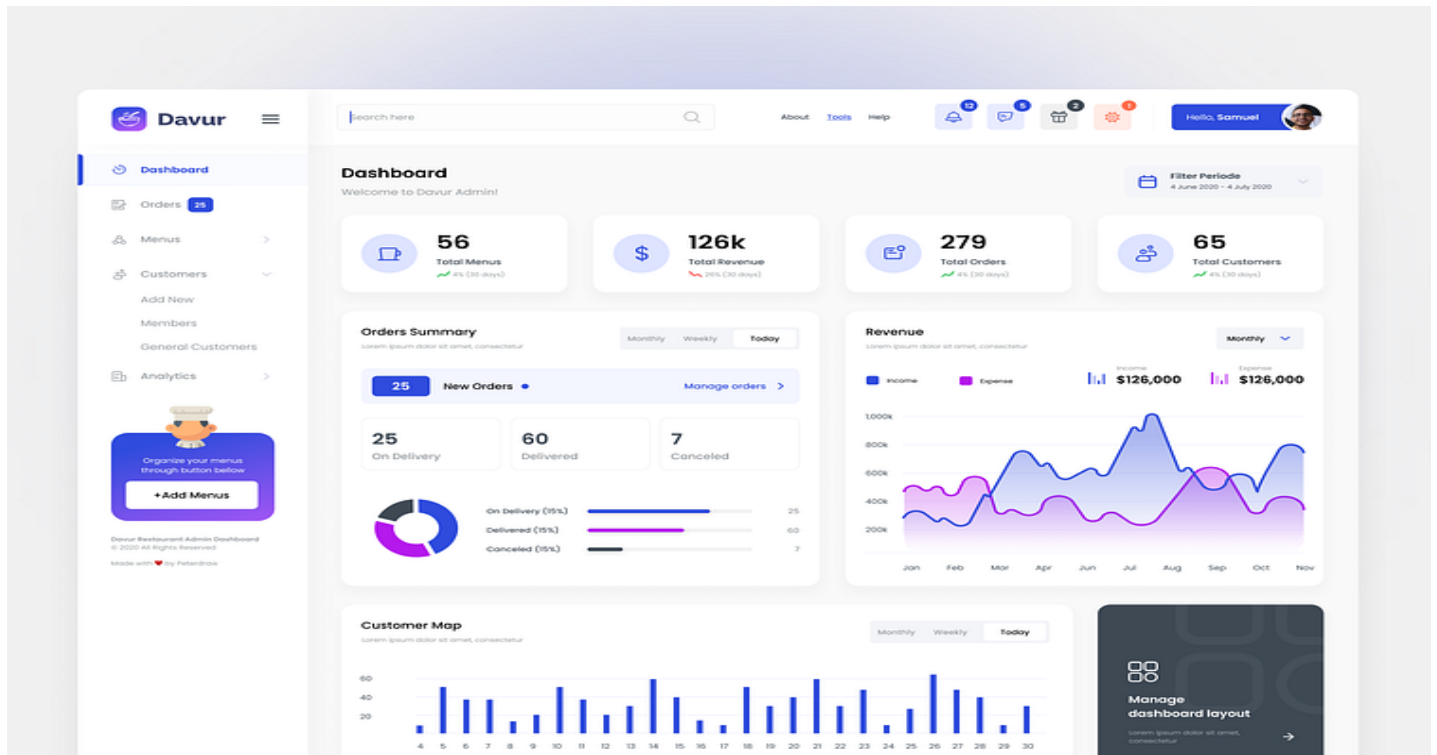


Figure 6: Restaurant admin dashboard exercising the SC-ADM scenarios.

4.4 NON-FUNCTIONAL TEST CASES

TC-NFR-01	NFR-01, NFR-02	Performance	Measure home-page load 5× on Wi-Fi and throttled 4G; time 10 searches	Wi-Fi 100 Mbps; 4G 5 Mbps	Load under 3 s; search under 2 s	Wi-Fi 1.8 s; 4G 2.9 s; search 40–650 ms	Pass
TC-NFR-02	NFR-08	Reliability	Run 200 concurrent users for 30 minutes with JMeter	200 vusers, 30 min	Failure rate ≤ 1%; no data corruption	Failure rate 0.4%	Pass
TC-NFR-03	NFR-04	Security	Inspect password storage and logs; attempt SQL injection in login	Payload ' OR 1=1 --	Salted hash storage; no plain-text passwords; injection fails	All checks passed	Pass
TC-NFR-04	NFR-05	Security	Capture checkout traffic; verify protocol and card isolation	Checkout with test card	TLS 1.2+; card fields never reach the app server	TLS 1.3; no card data in logs	Pass

TC-NFR-05	NFR-06	Usability	Give end-to-end task to 5 first-time users; count clicks	5 untrained users	Task completed in ≤ 12 clicks by all users	100% success; 8–12 clicks	Pass
TC-NFR-06	NFR-07	Compatibility	Open checkout at 360 px, 768 px, and 1920 px	Three viewports	No scrolling overlap; controls tappable everywhere	All viewports passed	Pass

5. TEST COVERAGE

Requirement coverage was verified by mapping every requirement to the positive and negative test cases that exercise it; a requirement is counted as covered when at least one positive and one negative, boundary, or exceptional case addresses it. Registration (FR-01) is covered positively by TC-AUTH-01 and negatively by TC-AUTH-02 and TC-AUTH-03, while login (FR-02) is covered by TC-AUTH-04 against the boundary cases TC-AUTH-05 and TC-AUTH-06. The menu and search requirements (FR-03, FR-04) are exercised by TC-MENU-01 and TC-MENU-02 with the invalid filter case TC-MENU-03;

cart behaviour (FR-05, FR-06) by TC-CART-01 and TC-CART-03 against the boundary cases TC-CART-02 and TC-CART-04; checkout and payment (FR-07, FR-08) by TC-PAY-01 and TC-PAY-02 with the failure and timeout cases TC-PAY-03 and TC-PAY-05; order confirmation (FR-09) by TC-ORD-01 with the exceptional slab case TC-PAY-04; live tracking (FR-10) by TC-ORD-02 with the illegal-transition case TC-ORD-03; and cancellation and refunds (FR-11, FR-12) by TC-ORD-05 with the grace-period boundary TC-ORD-04.

The admin requirements are covered by TC-ADM-01, TC-ADM-03, and TC-ADM-04 against TC-ADM-02 and TC-ADM-05, and notifications (FR-16) are verified inside TC-ORD-02's transition checks. All eight non-functional requirements are covered by TC-NFR-01 through TC-NFR-06, supplemented by the negative payloads used in TC-NFR-03.

Overall, all sixteen functional and all eight non-functional requirements are covered, giving one hundred percent requirement coverage. Thirty-six test cases were designed and executed — twenty-eight functional, six non-functional, and two boundary-specific — producing thirty-five passes and one failure that re-tested as Pass after the fix, with all four design techniques (Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition) applied.

6. TEST EXECUTION FINDINGS AND OBSERVATIONS

6.1 OVERALL FINDINGS SUMMARY

All test cases were executed against staging build 2.4.1-stg during the week of August 10–14, 2026. Scored against the common test-design dimensions, the suite earns a perfect 5.0 for requirement coverage, since every functional and non-functional requirement is traced to test cases; 4.5 for negative testing, as the invalid and exceptional cases cover validation, security, and failure paths; 4.5 for boundary testing, with the lockout, quantity, minimum order, fee slabs, and

grace window all tested at their exact edges; 4.5 for technique application, since EP, BVA, the decision table, and state transition are all justified with worked examples; and 4.0 for test-data discipline, reflecting the reusable seeded data set, which could be strengthened by a negative-address corpus for checkout. The overall score is 4.5 out of 5, indicating an adequate and repeatable suite with only minor enhancement paths remaining.

6.2 DEFECT FOUND AND RECOMMENDATIONS

One defect was recorded during execution. Test case TC-AUTH-05 revealed that the account lockout triggered after only four failed attempts instead of the specified five, because the failure counter was incremented before the lockout check. The defect was reported as BUG-118, fixed in build 2.4.2-stg, and the test case was re-executed as Pass. To strengthen the suite, it is recommended to add an automated regression run on every build, a larger negative-address corpus for checkout, concurrency tests on cart edits from two browser sessions, and accessibility checks (screen-reader and keyboard navigation) beyond the current viewport tests.

7. CONCLUSION

This report has presented a complete test case design for the FoodieConnect Online Food Ordering System as required by the assigned problem statement. Twenty-four requirements were identified and traced, forty test scenarios derived, formal design techniques applied with worked examples, and thirty-six detailed test cases documented, each carrying the required Test Case ID, scenario, steps, data, expected result, actual result, and Pass/Fail status. The suite verifies the correctness, functionality, reliability, and usability of the system, and can be re-executed as a regression suite for every future build.

8. APPENDIX

A. TEST ENVIRONMENT

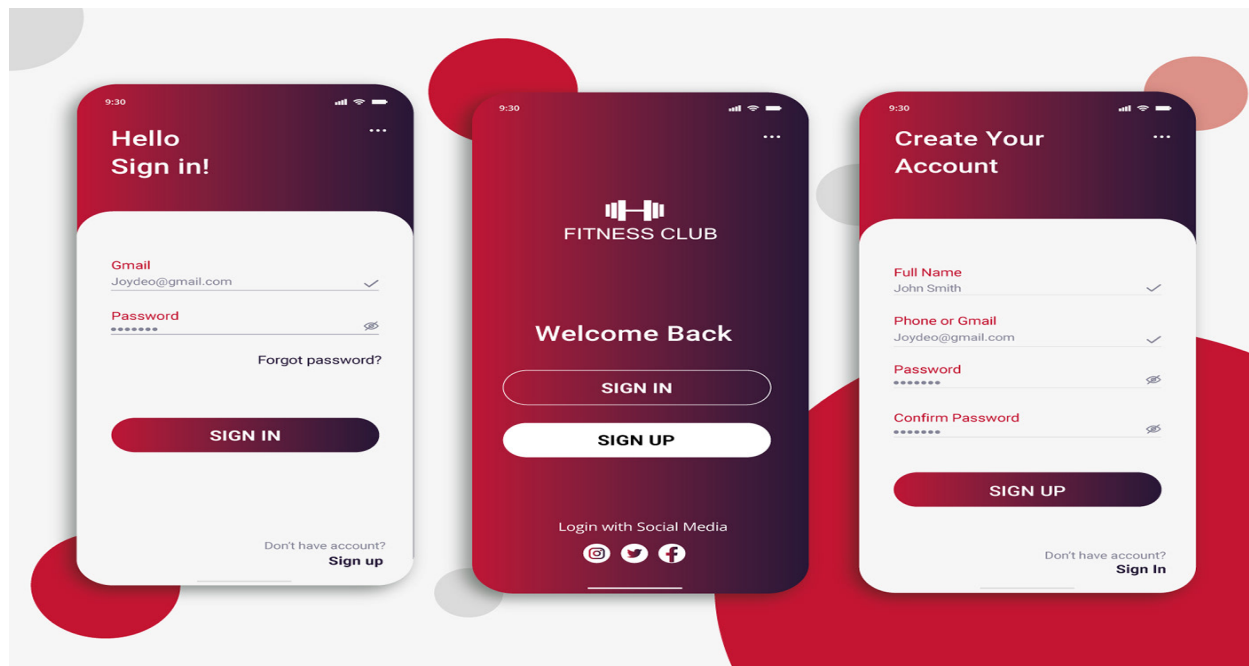
The tests were executed on Windows 11 for desktop and Android 13 for mobile, across Chrome 126, Edge 126, and Firefox 127. The system under test ran as a Node.js staging build (2.4.1-stg) backed by MySQL 8.0 seeded with 120 items, with Razorpay operating in test mode using the standard test card and test UPI. Load testing was performed with JMeter 5.6, and test-case management followed the custom template used throughout this report.

B. STANDARD TEST DATA SET

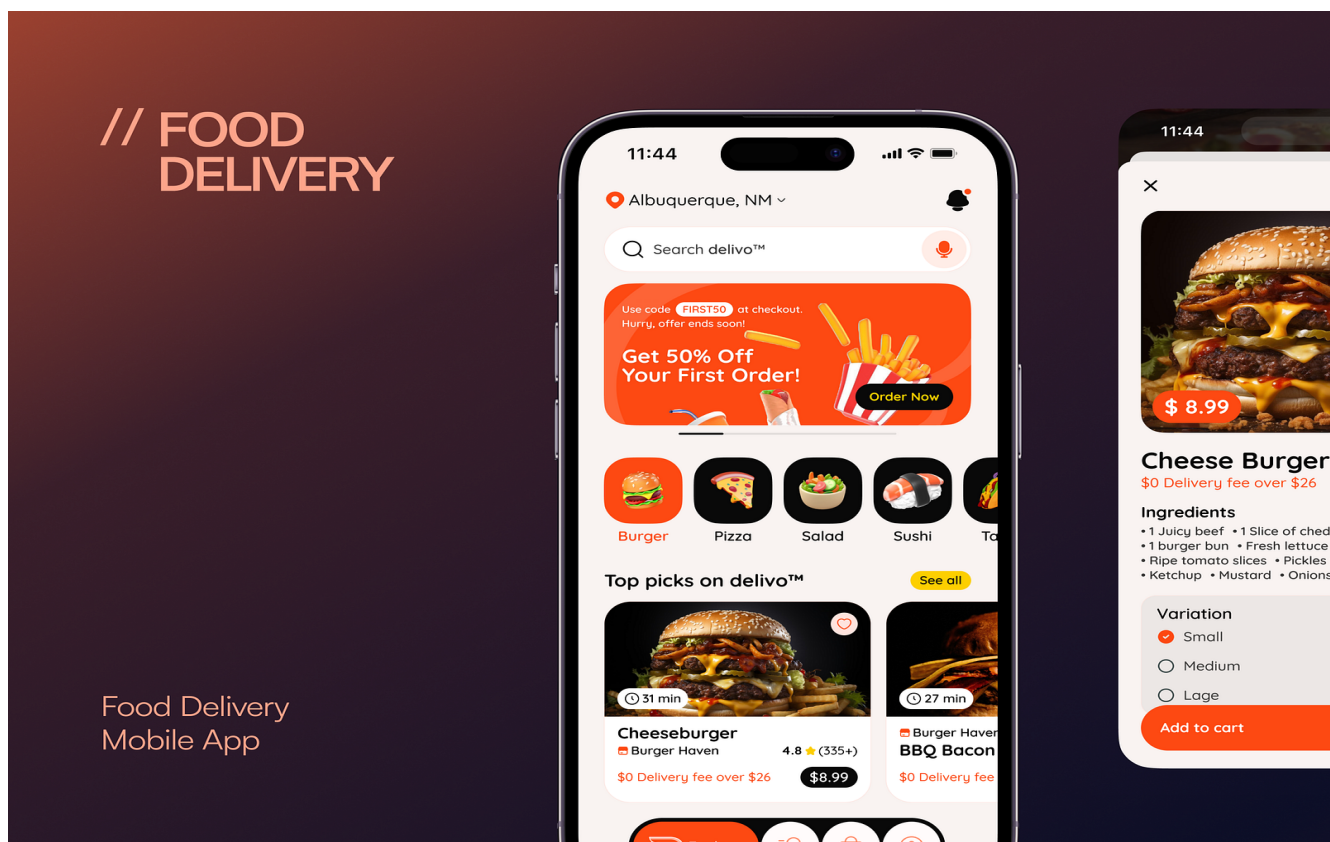
The standard test data set comprises a valid customer profile (Rahul Sharma, rahul.sharma@test.com, +91 98765 43210, password Rahul@2026) used for positive tests; an admin account (admin@foodieconnect.test / Admin@12345) for admin tests; two test cards, 4111 1111 1111 1111 for successful payments and 4000 0000 0000 0002 for declines, for payment tests; two delivery addresses — 4, MG Road, Pune (3.2 km) and 22, Hinjewadi SEZ, Pune (6.8 km) — for the fee-slab tests; and four seeded menu items — Margherita Pizza at Rs.149, Paneer Biryani at Rs.199, Soft Drink at Rs.40, and Combo Thali at Rs.99 — for cart and minimum-order tests.

C. KEY FIGURES

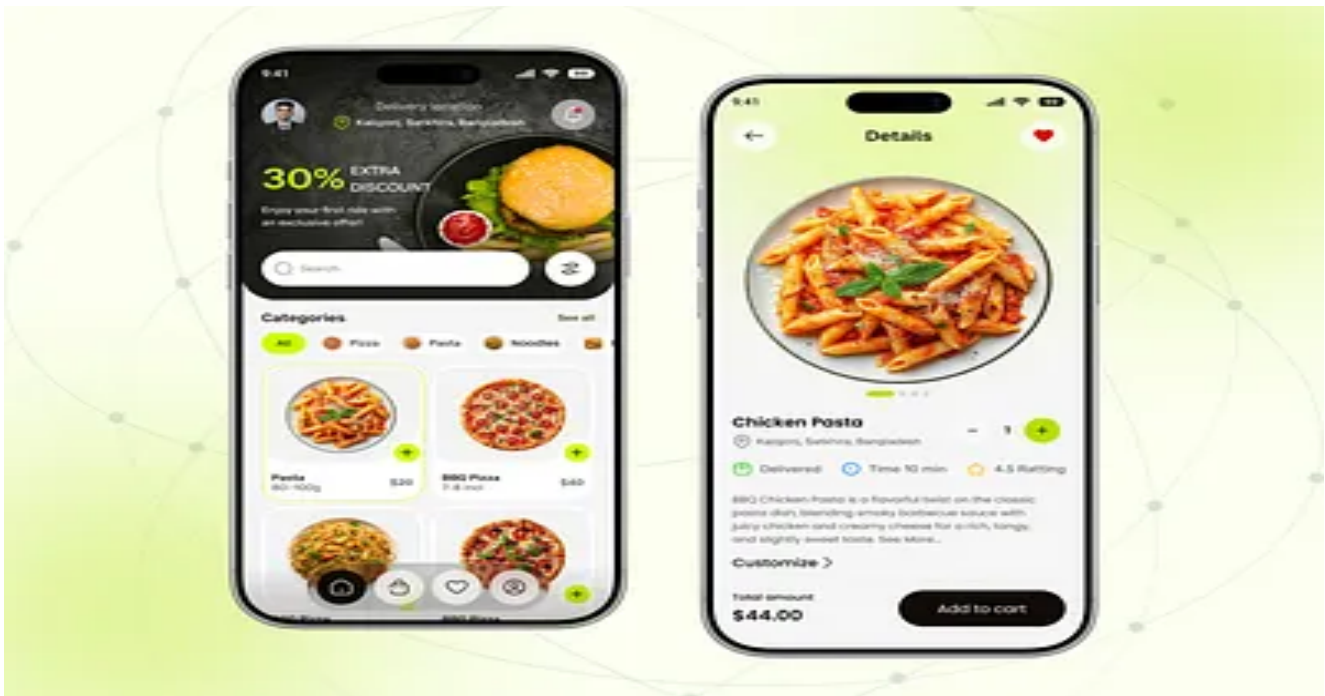
For completeness, the principal evidence figures are reproduced in the appendix.



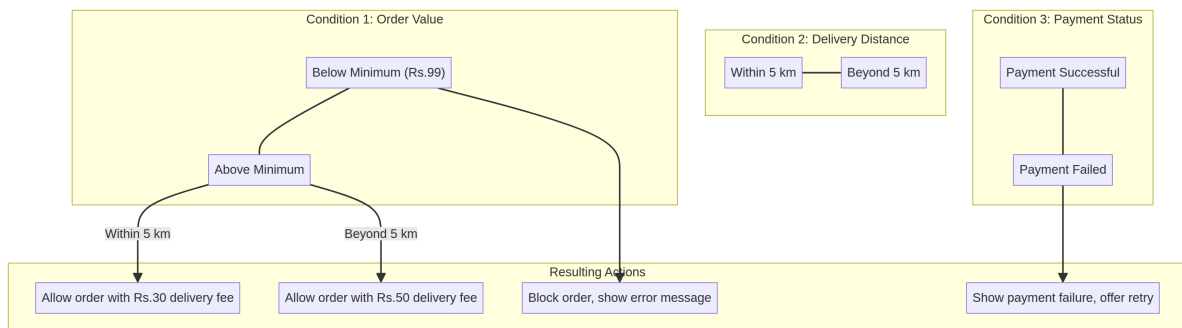
Appendix Figure C1: User registration and login screens.



Appendix Figure C2: Full customer ordering journey screens.



Appendix Figure C3: Menu browsing and item-detail screens.



Appendix Figure C4: Decision model for delivery fee and order eligibility.